

Differential Equations Lesson 06

Question 1:

Solve the differential equation $(x^3y - y^3 + 2)dx + \left(\frac{1}{4}x^4 - 3xy^2 + y\right)dy = 0$

Note: $\frac{\partial M(x,y)}{\partial y} = \frac{\partial}{\partial y}(x^3y - y^3 + 2) = x^3 - 3y^2$ and $\frac{\partial N(x,y)}{\partial x} = \frac{\partial}{\partial x}\left(\frac{1}{4}x^4 - 3xy^2 + y\right) = x^3 - 3y^2$

This means that the differential equation $(x^3y - y^3 + 2)dx + \left(\frac{1}{4}x^4 - 3xy^2 + y\right)dy = 0$ is an exact differential equation.

Hence, there exists a function $F(x, y)$ such that

$$F_x(x, y) = x^3y - y^3 + 2 \quad \text{and} \quad F_y(x, y) = \frac{1}{4}x^4 - 3xy^2 + y$$

a) $F(x, y) = \int F_x(x, y) \cdot dx = \int (x^3y - y^3 + 2) \cdot dx = \underline{\hspace{2cm}} + C(y)$

b) Use part (a) to find $F_y(x, y) = \underline{\hspace{2cm}} + C'(y)$

c) $C'(y) = \underline{\hspace{2cm}}$

d) $C(y) = \underline{\hspace{2cm}}$

e) The general solution of the differential equation $(x^3y - y^3 + 2)dx + \left(\frac{1}{4}x^4 - 3xy^2 + y\right)dy = 0$ is $\underline{\hspace{2cm}}$

Question 2:

Solve the differential equation $(3x^2y + 8y^3)dx + (x^3 + 24y^2x)dy = 0$

Note: $\frac{\partial M(x,y)}{\partial y} = \frac{\partial}{\partial y}(3x^2y + 8y^3) = 3x^2 + 24y^2$ and $\frac{\partial N(x,y)}{\partial x} = \frac{\partial}{\partial x}(x^3 + 24y^2x) = 3x^2 + 24y^2$

This means that the differential equation $(3x^2y + 8y^3)dx + (x^3 + 24y^2x)dy = 0$ is an exact differential equation.

Hence, there exists a function $F(x,y)$ such that

$$F_x(x,y) = 3x^2y + 8y^3 \quad \text{and} \quad F_y(x,y) = x^3 + 24y^2x$$

a) $F(x,y) = \int F_x(x,y) \cdot dx = \int (3x^2y + 8y^3) \cdot dx = \underline{\hspace{2cm}} + C(y)$

b) Use part (a) to find $F_y(x,y) = \underline{\hspace{2cm}} + C'(y)$

c) $C'(y) = \underline{\hspace{2cm}}$

d) $C(y) = \underline{\hspace{2cm}}$

e) The general solution of the differential equation $(3x^2y + 8y^3)dx + (x^3 + 24y^2x)dy = 0$ is $\underline{\hspace{2cm}}$

Question 3:

Solve the differential equation $(x^3 + 9x^2y)dx + \left(3 + \frac{1}{4}x^4 + 3x^3\right)dy = 0$

Note: $\frac{\partial M(x,y)}{\partial y} = \frac{\partial}{\partial y}(x^3y + 9x^2y) = x^3 + 9x^2$ and $\frac{\partial N(x,y)}{\partial x} = \frac{\partial}{\partial x}\left(3 + \frac{1}{4}x^4 + 3x^3\right) = x^3 + 9x^2$

This means that the differential equation $(x^3 + 9x^2y)dx + \left(3 + \frac{1}{4}x^4 + 3x^3\right)dy = 0$ is an exact differential equation.

Hence, there exists a function $F(x,y)$ such that

$$F_x(x,y) = x^3y + 9x^2y \quad \text{and} \quad F_y(x,y) = 3 + \frac{1}{4}x^4 + 3x^3$$

a) $F(x,y) = \int F_x(x,y) \cdot dx = \int (x^3y + 9x^2y) \cdot dx = \underline{\hspace{2cm}} + C(y)$

b) Use part (a) to find $F_y(x,y) = \underline{\hspace{2cm}} + C'(y)$

c) $C'(y) = \underline{\hspace{2cm}}$

d) $C(y) = \underline{\hspace{2cm}}$

e) The general solution of the differential equation $(x^3 + 9x^2y)dx + \left(3 + \frac{1}{4}x^4 + 3x^3\right)dy = 0$ is $\underline{\hspace{2cm}}$

Question 4:

Solve the differential equation $\left(\frac{5y}{x} + 5x^2\right)dx + (5\ln x - 4y)dy = 0$ where $x > 0$

Note: $\frac{\partial M(x, y)}{\partial y} = \frac{\partial}{\partial y}\left(\frac{5y}{x} + 5x^2\right) = \frac{5}{x}$ and $\frac{\partial N(x, y)}{\partial x} = \frac{\partial}{\partial x}(5\ln x - 4y) = 5 \cdot \frac{1}{x} = \frac{5}{x}$

This means that the differential equation $\left(\frac{5y}{x} + 5x^2\right)dx + (5\ln x - 4y)dy = 0$ is an exact differential equation.

Hence, there exists a function $F(x, y)$ such that

$$F_x(x, y) = \frac{5y}{x} + 5x^2 \quad \text{and} \quad F_y(x, y) = 5\ln x - 4y$$

a) $F(x, y) = \int F_x(x, y) \cdot dx = \int \left(\frac{5y}{x} + 5x^2\right) \cdot dx = \int \left(5y \cdot \frac{1}{x} + 5x^2\right) \cdot dx = \underline{\hspace{2cm}} + C(y)$

b) Use part (a) to find $F_y(x, y) = \underline{\hspace{2cm}} + C'(y)$

c) $C'(y) = \underline{\hspace{2cm}}$

d) $C(y) = \underline{\hspace{2cm}}$

e) The general solution of the differential equation $\left(\frac{5y}{x} + 5x^2\right)dx + (5\ln x - 4y)dy = 0$ is $\underline{\hspace{2cm}}$

Question 5:

Solve the differential equation $(4x \cdot \sin y - y \sin x) dx + (2x^2 \cos y + \cos x) dy = 0$

Note: $\frac{\partial M(x, y)}{\partial y} = \frac{\partial}{\partial y}(4x \cdot \sin y - y \sin x) = 4x \cdot \cos y - \sin x$

$$\frac{\partial N(x, y)}{\partial x} = \frac{\partial}{\partial x}(2x^2 \cos y + \cos x) = 4x \cos y - \sin x$$

This means that the differential equation $(4x \cdot \sin y - y \sin x) dx + (2x^2 \cos y + \cos x) dy = 0$ is an exact differential equation.

Then there exists a function $F(x, y)$ such that

$$F_x(x, y) = 4x \cdot \sin y - y \sin x \quad \text{and} \quad F_y(x, y) = 2x^2 \cos y + \cos x$$

a) $F(x, y) = \int F_x(x, y) \cdot dx = \int (4x \cdot \sin y - y \sin x) \cdot dx = \underline{\hspace{2cm}} + C(y)$

b) Use part (a) to find $F_y(x, y) = \underline{\hspace{2cm}} + C'(y)$

c) $C'(y) = \underline{\hspace{2cm}}$

d) $C(y) = \underline{\hspace{2cm}}$

e) The general solution of the differential equation $(4x \cdot \sin y - y \sin x) dx + (2x^2 \cos y + \cos x) dy = 0$ is $\underline{\hspace{2cm}}$

